

Course Outline: Digital Logic and Design

1. Introduction to Digital Logic

- Introduction to Digital Logic
- Analog Data vs. Digital Data
- Analog Systems vs. Digital Systems
- How Digital Systems Work
- Advantages of Digital Over Analog Systems
- Number Systems (Binary, Octal, Decimal, Hexadecimal)
- Conversion between Number Systems
- Binary Arithmetic and Complements (1's and 2's Complement)

2. Boolean Algebra and Logic Gates

- Logic Gates
- Classification of Logic Gates
 - Basic Gates (AND, OR, NOT)
 - Universal Gates (NAND, NOR)
 - Special Purpose Gates (XOR and XNOR)
- Boolean Algebra
- Boolean Algebra Laws and Theorems
- De Morgan's Theorem
- Duality Theorem
- Boolean Expression Simplification
- Applications of Boolean Algebra

3. Logic Simplification and Karnaugh Maps (K-Maps)

- Minterms
- Maxterms
- Boolean Function Representation
 - SOP (Sum of Products)
 - POS (Product of Sums)
- Conversion from SOP to POS
- Conversion from POS to SOP
- Canonical Form
- Standard Form
- Karnaugh Map (K-Map)
- Simplification using K-Map
 - K-Map for SOP
 - K-Map for POS

4. Combinational Logic Circuits

- Combinational Logic Circuits
- Adders (Half Adders, Full Adders)
- Subtractors (Half Subtractors, Full Subtractors)
- Multiplexers (MUX)
 - Advantages and Disadvantages
- DE Multiplexers (DEMUX)
 - Advantages and Disadvantages
- Encoders
 - 3x8 Encoder (Truth Table and Implementation)
- Decoders
 - 8x3 Decoder (Truth Table and Implementation)
- Parity Checkers
 - Working
 - Types

- Parity Checkers
 - Working
 - Types
- Code Converters
 - BCD to 7-Segment
 - Binary to Gray Code

5. Sequential Logic Circuits

- Synchronous and Asynchronous Sequential Circuits
- Latches
 - Types of Latches
 - Advantages and disadvantages
- Flip-Flops
 - Types of Flip-Flops
 - Advantages and disadvantages
- Counters
 - Asynchronous and Synchronous Counters
 - Applications
 - Advantages and disadvantages
- Registers and Shift Registers
 - Types of Registers
 - Applications
 - Advantages and disadvantages